

PROG 10 :

Step 1 : create 2 files Deployment.yaml and service.yaml

```
rv24mc028_chethanacs@Elite:~/DEVops/prog10$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
rv24mc028_chethanacs@Elite:~/DEVops/prog10$ █
```

```
rv24mc028_chethanacs@Elite:~/DEVops/prog10$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
rv24mc028_chethanacs@Elite:~/DEVops/prog10$
```

Step 2 : apply for both

```
rv24mc028_chethanacs@Elite:~/DEVops/prog10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
rv24mc028_chethanacs@Elite:~/DEVops/prog10$ kubectl apply -f service.yaml
service/nginx-service created
```

Step 3 : get all the pods

```
rv24mc028_chethanacs@Elite:~/DEVops/prog10$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
backup-job-29445750-ffmqv	0/1	Completed	0	6d23h
backup-job-29455775-dn5mq	0/1	Completed	0	5m7s
backup-job-29455780-64gg9	0/1	Completed	0	4m34s
my-cron-29455782-l5xbm	0/1	Completed	0	2m34s
my-cron-29455783-6nnpd	0/1	Completed	0	94s
my-cron-29455784-l75xl	0/1	Completed	0	34s
nginx-deployment-6f9664446b-84bn4	1/1	Running	0	28s
nginx-deployment-6f9664446b-f4nz2	1/1	Running	0	28s

Step 4 : check all the nodes and services

```
rv24mc028_chethanacs@Elite:~/DEVops/prog10$ kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
minikube	Ready	control-plane	37d	v1.34.0

```
rv24mc028_chethanacs@Elite:~/DEVops/prog10$ kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	6d23h
nginx-service	NodePort	10.97.210.179	<none>	80:32081/TCP	19s

Step 5 : check all logs of a pod

```
rv24mc028_chethanacs@Elite:~/DEVops/prog10$ kubectl logs nginx-deployment-6f9664446b-84bn4
```

```
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/01/02 09:44:12 [notice] 1#1: using the "epoll" event method
2026/01/02 09:44:12 [notice] 1#1: nginx/1.29.4
2026/01/02 09:44:12 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/02 09:44:12 [notice] 1#1: OS: Linux 6.8.0-87-generic
2026/01/02 09:44:12 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/02 09:44:12 [notice] 1#1: start worker processes
2026/01/02 09:44:12 [notice] 1#1: start worker process 29
2026/01/02 09:44:12 [notice] 1#1: start worker process 30
2026/01/02 09:44:12 [notice] 1#1: start worker process 31
2026/01/02 09:44:12 [notice] 1#1: start worker process 32
2026/01/02 09:44:12 [notice] 1#1: start worker process 33
2026/01/02 09:44:12 [notice] 1#1: start worker process 34
2026/01/02 09:44:12 [notice] 1#1: start worker process 35
2026/01/02 09:44:12 [notice] 1#1: start worker process 36
```

step 6 : describe the pod

```

rv24mc028_chethanacs@Elite:~/DEVops/prog10$ kubectl describe pod nginx-deployment-6f9664446b-84bn4
Name:          nginx-deployment-6f9664446b-84bn4
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Fri, 02 Jan 2026 15:14:06 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.1.11
IPs:
  IP:          10.244.1.11
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://97a22c9dd763d9dd2a7d33f260aa1bfd2d2917df4328f819d39bf4161bd54493

```

Step 7 : check the services

```

rv24mc028_chethanacs@Elite:~/DEVops/prog10$ minikube service nginx-service

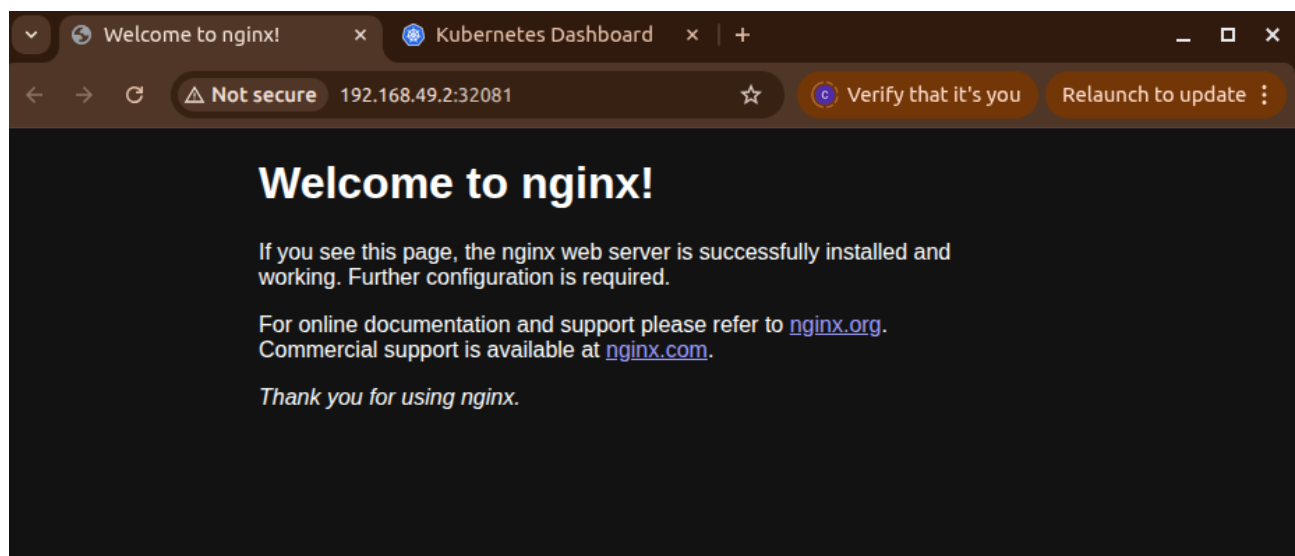
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:32081

```

🚀 Opening service default/nginx-service in default browser...

```



Step 8 : check the dashboard

```

rv24mc028_chethanacs@Elite:~/DEVops/prog10$ minikube dashboard
Enabling dashboard ...
  ■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
  ■ Using image docker.io/kubernetesui/dashboard:v2.7.0
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🤔 Verifying dashboard health ...
🚀 Launching proxy ...
🤔 Verifying proxy health ...
🚀 Opening http://127.0.0.1:32973/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:443
Opening in existing browser session.

```

